

DANIEL M. KROLIKOWSKI

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WORK EXPERIENCE

Postdoctoral Research Associate

September 2022 - Present

→ *Steward Observatory, University of Arizona*

I am a software and science team member for the extreme precision radial velocity spectrometers NEID and HPF. I support operations, and develop features and address issues for their data pipelines.

★ I am an Architect on the NEID Team, denoting significant contribution to the project.

EDUCATION

The University of Texas at Austin

August 2016 - August 2022

Ph.D. Astronomy

Exploring the Formation and Evolutionary Pathways of Young Stars and Planetary Systems at High Precision

Advisor: Dr. Adam Kraus

State University of New York College at Geneseo

September 2012 - May 2016

B.A. Physics, B.A. Mathematics

COMPUTER SKILLS

Computer Languages: Python, Bash, SQL/ADQL, Markdown, MATLAB, C

Tools and Software: git/GitHub, L^AT_EX, Slurm/HPC, Sphinx, astropy, JAX, IRAF, Mathematica

Data File Formats: FITS, CSV, HDF5, Astropy Table

SOFTWARE PROJECTS AND CONTRIBUTIONS

NEID Data Reduction Pipeline

<https://neid.ipac.caltech.edu/docs/NEID-DRP/>

I am an active developer on the NEID SpecSoft team. I rewrote and upgraded the pipeline's telluric model generation, refactored and overhauled the complex wavelength calibration module, fixed bugs, and streamlined operations to ensure high quality data processing. I contributed to DRP versions 1.2 – 1.5.

EPRV Community Data Format Standard and Instrument Data Translators

<https://github.com/EPRV-RCN/RVData>

A package to translate data from native EPRV instrument data formats to a community standardized FITS data format. This is in collaboration with EPRV astronomers and instrument teams around the world. I helped to define the standard data format and am a primary developer of the NEID translator.

Tull Coudé Spectrograph Reduction and Analysis Pipeline

https://github.com/dkrolikowski/tull.coude_reduction

A python reduction and analysis pipeline for the Tull coudé spectrograph on McDonald Observatory's 2.7-m telescope. I am the sole developer and have maintained the pipeline since 2017.

Packages with minor contributions

→ **hpfserveral**: Measuring radial velocities for HPF using template matching (adapted from SERVAL).

Contributions: added telluric correction of spectra prior to analysis. (<https://github.com/gummiks/hpfserveral>)

→ **SAPHIRES**: Measuring radial velocities from stellar spectra using broadening functions.

Contributions: testing, feedback on input/output framework. (<https://github.com/tofflemire/saphires>)

→ **Comove**: Searching for co-moving companions to a given star in *Gaia* and compiling other data.

Contributions: testing, feedback on plotting and data output. (<https://github.com/adamkraus/Comove>)

→ **muler**: Data structure to handle and analyze echelle spectra in python (specifically HPF).

Contributions: Sky subtraction and HPF data best practices. (<https://github.com/OttoStruve/muler>)

PUBLICATIONS

First Author

2. *The Strength and Variability of the Helium 10830 Å Triplet in Young Stars with Implications for Exosphere Detection*
Daniel Krolkowski, Adam Kraus, Benjamin Tofflemire, Caroline Morley, Andrew Mann, Andrew Vanderburg, AJ, 167, 79
1. *Gaia EDR3 Reveals the Substructure and Complicated Star Formation History of the Greater Taurus-Auriga Star-forming Complex*
Daniel Krolkowski, Adam Kraus, Aaron Rizzuto, 2021, AJ, 162, 110

In prep.

3. *Measuring the Mass of a 3 Myr Old Jupiter-radius Planet, TIDYE-1 b*
Daniel Krolkowski, Madyson Barber, Andrew Mann, Adam Kraus, et al. In prep.
2. *The NEID Data Reduction Pipeline I: Overview*
Chad Bender, **Daniel Krolkowski**, et al., In prep.
1. *The Discovery of Four Lithium Rich Red Giants in a Single Open Cluster, NGC 2158*
Daniel Krolkowski, Aaron Steinhauer, Constantine Deliyannis, Barbara Anthony-Twarog, Bruce Twarog. In prep.

Co-author

38. *Confirmation of the Low-eccentricity Warm Jupiter HD 278555 b with NEID*
James Ball, Arvind Gupta, Diana Dragomir, ..., **Daniel Krolkowski**, et al., 2026, AJ, 172, 158
37. *Searching for GEMS: TOI-5916 b & TOI-6158 b are two Saturn-density planets orbiting M2 dwarfs*
Shane O'Brien, Amber Wong, Te Han, ..., **Daniel Krolkowski**, et al., 2026, AJ, 172, 117
36. *Searching for GEMS: Three warm Saturns and a super-Jupiter orbiting four early M-dwarfs*
Pranav Premnath, Paul Robertson, Shubham Kanodia, ..., **Daniel Krolkowski**, et al., 2026, AJ, 172, 92
35. *A Revised Mass and Period for the Habitable Zone super-Earth GJ 3378b: A Planet Straddling the Cosmic Shoreline*
Paul Robertson, Michael Endl, William Cochran, ..., **Daniel Krolkowski**, et al., 2026, ApJ, 1005, 32
34. *Improving the Precision of Line-by-Line Radial Velocities: A Data-Driven Iterative Algorithm for Spectral Line Selection*
Kanishk Pandey, Eric Ford, Joseph Salzer, ..., **Daniel Krolkowski**, et al., 2026, arXiv:2606.29552, Submitted AAS Journals
33. *The NEID Earth Twin Survey. IV. Confirming an 89 d, $m \sin i = 10 M_{\oplus}$ Planet Orbiting a Nearby Sun-like Star*
Mark Giovannazzi, Evan Fitzmaurice, Arvind Gupta, ..., **Daniel Krolkowski**, et al., 2026, AJ, 171, 268
32. *TOI-1259Ab: A Warm Jupiter Orbiting a K-dwarf White-Dwarf Binary is on a Well-aligned Orbit*
Hugo Veldhuis, Juan Espinoza-Retamal, Gudmundur Stefansson, ..., **Daniel Krolkowski**, et al., 2025, A&A, 708, 272
31. *HET/HPF observations of Helium in warm, hot, and ultra-hot Jupiters*
Jaume Orell-Miquel, Kyra Sampson, Caroline Morley, ..., **Daniel Krolkowski**, et al., 2026, AJ, 171, 194

30. *Stellar and orbital characterization of three low mass M dwarf binary stars with dynamical spectroscopy from the Habitable Zone Planet*
Suhani Surana, Chad Bender, Caleb Cañas, **Daniel Krolikowski**, et al., 2026, AJ, 171, 193
29. *Searching for GEMS: The Occurrence of Giant Planets orbiting M-dwarfs within 100 pc*
Rowen Glusman, Caleb Cañas, Shubham Kanodia, ..., **Daniel Krolikowski**, et al., 2026, AJ, 171, 146
28. *An Earth-Sized Planet in a 5.4h Orbit Around a Nearby K dwarf*
Kaya Taş, Gudmundur Stefansson, Syarief Fariz, ..., **Daniel Krolikowski**, et al., 2025, A&A, 707, 238
27. *Spectroscopic Characterization of LOFAR Radio-emitting M dwarfs*
Elise Koo, Gudmundur Stefansson, Robert Kavanagh, ..., **Daniel Krolikowski**, et al., 2026, A&A, 706, 98
26. *A Simple One-free-parameter Model of the Solar Chromosphere Explains Solar Chromospheric Line Variations Measured by NEID at WIYN during the 2023 October Solar Eclipse*
Jason Wright, Elizabeth Gonzalez, ..., **Daniel Krolikowski**, et al., 2026, RNAAS, 10, 14
25. *Stellar Obliquities of Young Systems, Atmospheres Undergoing Contraction and Escape (SOYSAUCE): A Likely Aligned Orbit for the 3 Myr Planet TIDYE-1 b*
Madyson Barber, Andrew Mann, Marshall Johnson, ..., **Daniel Krolikowski**, et al., 2025, ApJL, 994, L55
24. *Order-by-order Modeling of Exoplanet Radial Velocity Data*
Zachary Langford, Cullen Blake, Samuel Halverson, ..., **Daniel Krolikowski**, et al., 2025, PASP, 137, 4503
23. *Discovery of a Nearby Habitable Zone Super-Earth Candidate Amenable to Direct Imaging*
Corey Beard, Paul Robertson, Jack Lubin, ..., **Daniel Krolikowski**, et al., 2025, AJ, 170, 279
22. *The NEID Earth Twin Survey. III. Survey Performance After Three Years on Sky*
Arvind Gupta, Evan Fitzmaurice, Suvrath Mahadevan, ..., **Daniel Krolikowski**, et al., 2025, AJ, 170, 264
21. *Searching for GEMS: TOI-7149 b an Inflated Giant Planet causing a 12% Transit of a Fully Convective M-dwarf*
Shubham Kanodia, Caleb Cañas, Suvrath Mahadevan, ..., **Daniel Krolikowski**, et al., 2025, AJ, 170, 203
20. *Searching for GEMS: Confirmation of TOI-5573b, a Cool, Saturn-like Planet Orbiting An M-dwarf*
Rachel Fernandes, Shubham Kanodia, Megan Delamer, ..., **Daniel Krolikowski**, et al., 2025, AJ, 170, 55
19. *NEIDSpecMatch: stellar parameter estimation with NEID spectra using an empirical library*
Te Han, Paul Robertson, Caleb Cañas, ..., **Daniel Krolikowski**, et al., 2025, RNAAS, 9, 63
18. *Multi-Observatory Study of Young Stellar Energetic Flares (MORYSEF): No Evidence For Abnormally Strong Stellar Magnetic Fields After Powerful X-ray Flares*
Konstantin Getman, Oleg Kochukhov, Joe Ninan, ..., **Daniel Krolikowski**, 2025, ApJ, 980, 57
17. *Gaia-4b and 5b: Radial Velocity Confirmation of Gaia Astrometric Orbital Solutions Reveal a Massive Planet and a Brown Dwarf Orbiting Low-mass Stars*
Gudmundur Stefansson, Suvrath Mahadevan, Joshua Winn, ..., **Daniel Krolikowski**, et al., 2025, AJ, 169, 107
16. *Jitter Across 15 Years: Leveraging Precise Photometry from Kepler and TESS to Extract Exoplanets from Radial Velocity Time Series*

- Corey Beard, Paul Robertson, Jack Lubin, ..., **Daniel Krolkowski**, et al., 2025, AJ, 169, 92
15. *The NEID Earth Twin Survey. I. Confirmation of a 31 Day Planet Orbiting HD 86728*
Arvind Gupta, Jacob Luhn, Jason Wright, ..., **Daniel Krolkowski**, et al., 2025, AJ, 169, 1
 14. *A giant planet transiting a 3-Myr protostar with a misaligned disk*
Madyson Barber, Andrew Mann, Andrew Vanderburg, **Daniel Krolkowski**, et al., 2024, Nature, 635, 574
 13. *SPYGLASS. V. Spatially and Temporally Structured Star-forming Environments in the Cepheus-Hercules Complex*
Ronan Kerr, Adam Kraus, **Daniel Krolkowski**, et al., 2024, ApJ, 975, 99
 12. *A Large and Variable Leading Tail of Helium in a Hot Saturn Undergoing Runaway Inflation*
Michael Gully-Santiago, Caroline Morley, Jessica Luna, ..., **Daniel Krolkowski**, et al., 2024, AJ, 167, 142
 11. *Giant tidal tails of helium escaping the hot Jupiter HAT-P-32 b*
Zhoujian Zhang, Caroline Morley, Michael Gully-Santiago, ..., **Daniel Krolkowski**, et al., 2023, Science Advances, 9, 23
 10. *TOI-4562 b: A highly eccentric temperate Jupiter analog orbiting a young field star*
Alexis Heitzmann, George Zhou, Samuel Quinn, ..., **Daniel Krolkowski**, et al., 2023, AJ, 165, 121
 9. *SPYGLASS III: The Fornax-Horologium Association and its Traceback History within the Austral Complex*
Ronan Kerr, Adam Kraus, Simon Murphy, **Daniel Krolkowski**, et al., 2022, ApJ, 941, 143
 8. *SPYGLASS II: The Multigenerational and Multiorigin Star Formation History of Cepheus Far North*
Ronan Kerr, Adam Kraus, Simon Murphy, **Daniel Krolkowski**, et al., 2022, ApJ, 941, 49
 7. *TESS Hunt for Young and Maturing Exoplanets (THYME) VII: Membership, rotation, and lithium in the young cluster Group-X and a new young exoplanet*
Elisabeth Newton, Rayna Rampalli, Adam Kraus, ..., **Daniel Krolkowski**, et al., 2022, AJ, 164, 115
 6. *Transit Hunt for Young and Maturing Exoplanets (THYME) VIII: a Pleiades-age association harboring two transiting planetary systems from Kepler*
Madyson Barber, Andrew Mann, Jonathan Bush, ..., **Daniel Krolkowski**, et al., 2022, AJ, 164, 88
 5. *Astronomical échelle spectroscopy data analysis with muler*
Michael Gully-Santiago, Jessica Luna, Caroline Morley, , ..., **Daniel Krolkowski**, 2022, JOSS, 7, 73
 4. *A 2+1+1 quadruple star system containing the most eccentric, low-mass, short-period, eclipsing binary known*
Eunkyu Han, Saul Rappaport, Andrew Vanderburg, ..., **Daniel Krolkowski**, et al., 2022, MNRAS, 510, 2448
 3. *NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star*
Jiayin Dong, Chelsea Huang, George Zhou, ..., **Daniel Krolkowski**, et al., 2022, ApJL, 926, L7
 2. *Close Companions around Young Stars*
Marina Kounkel, Kevin Covey, Maxwell Moe, ..., **Daniel Krolkowski**, et al., 2019, AJ, 157, 196

1. *Zodiacal Exoplanets in Time VIII. A Two-planet System in Praesepe from K2 Campaign 16* Aaron Rizzuto, Andrew Vanderburg, Andrew Mann, ..., **Daniel Krolkowski**, 2018, AJ, 156, 195

PRESENTATIONS

Invited Presentations

5. *Measuring the Mass of the Youngest Transiting Planet Amidst Extreme Stellar Noise*. Caltech/IPAC Lunch Seminar, March 11, 2026
4. *The Exoplanet Radial Velocity Frontier: Pushing Precision Limits and Exploring Young Planets with NEID and HPF*. SUNY Geneseo Department of Physics Colloquium, October 9, 2025
3. *Forward Modeling Telluric Absorption with a Variable Line Spread Function using NEID and HPF*. EPRV Research Coordination Network Colloquium, September 26, 2024
2. *Gaia EDR3 Reveals the Complicated Substructure and Star Formation History of Taurus*. University of Hawaii Star and Exoplanet Seminar, November 8, 2021
1. *Illuminating the Physics of Star and Planet Formation with the Nearest Newly-Formed Stars*. McDonald Observatory Board of Visitors Meeting, July 14, 2018

Contributed Talks

8. *Measuring the Mass of the Youngest Transiting Planet Amidst Extreme Stellar Noise*. Cool Stars 23 Conference: Young Stars with Evolving Planetary Systems Splinter Session, June 2026
7. *Measuring the Mass of the Youngest Transiting Planet Amidst Extreme Stellar Noise*. NOIRLab FLASH Talk, March 20, 2026
6. *NEID's Wavelength Calibration: Algorithms and Insights*. Wavelength calibration for high-resolution spectrographs workshop, May 2025
5. *The Helium NIR Triplet in Young Stars: Stellar Activity and Effects on Exosphere Detection*. Know thy Star Know thy Planet 2 Conference, February 2025
4. *Forward Modeling Telluric Absorption in High-Resolution Stellar Spectra*. 17th HITRAN Conference, June 2024
3. *Variability of the NIR helium line in young stars, and implications for detecting the evaporating atmospheres of young exoplanets*. Exoplanets IV Conference, May 2022
2. *Probing the Formation and Evolution of Young Planetary Systems with HPF*. THYME (TESS Hunt for Young and Maturing Exoplanets) Collaboration Meeting, December 2020
1. *Multiple Stellar Populations in Taurus and their 6D Structure*. GMT Science Meeting, September 2018

Contributed Posters

12. *Stellar Activity at the Youngest Ages: A Multi-Wavelength Campaign of the 3 Myr Old Planet Host TIDYE-1*. Cool Stars 23 Conference, June 2026
11. *Forward Modeling Telluric Absorption with a Variable Line Spread Function for NEID and HPF*. Extreme Precision Radial Velocity 6 Conference, July 2025
10. *The Outer Orbital Architectures of Young Transiting Planet Systems*. Extreme Solar Systems V Conference, March 2024
9. *The complex temporal and chromatic jitter signal of the M-dwarf EV Lac with multi-wavelength precision RVs*. Extreme Precision Radial Velocity 5 Conference, March 2023

8. *The Helium 10830 Å line in young stars: stellar activity and implications for exosphere detection.* Cool Stars 21 Conference, July 2022
7. *Precision RVs and He Spectroscopy of Young Planet Hosts with the Habitable-zone Planet Finder.* Extreme Solar Systems IV Conference, August 2019
6. *A Spectroscopic and Kinematic Survey of the Taurus-Auriga Star Forming Complex.* Cool Stars 20 Conference, July 2018
5. *A Spectroscopic and Kinematic Survey of the Taurus-Auriga Star Forming Complex.* Star and Planet Formation in the Southwest 2 Conference, March 2018
4. *WIYN Open Cluster Study: Lithium in Red Giants of the Open Cluster NGC 2158.* Council on Undergraduate Research: Posters on the Hill, April 2016
3. *WIYN Open Cluster Study: Lithium in Red Giants of the Open Cluster NGC 2158.* 227th AAS Meeting, January 2016
2. *Detecting Traces of Life in the Plume of Enceladus.* 225th AAS Meeting, January 2015
1. *WIYN Open Cluster Study: Li in the Open Cluster NGC 6811.* 225th AAS Meeting, January 2015

Public and Outreach Presentations

5. *Tracking Stellar Tortoises or: Finding Planets with Extremely Precise Speedometers.* Astronomy on Tap Grand Canyon, July 3, 2026
4. *Earth's analogue in the unlikely place, or why Titan is the coolest moon.* Astronomy on Tap ATX, September 17, 2019
3. *What the Taurus Star Forming Region Can Tell Us About Star and Planet Formation.* McDonald Observatory Special Viewing Night, November 1, 2017
2. *The Cosmic Importance of Lithium and the Mystery of Four Lithium Rich Stars in a Single Cluster.* Austin Astronomical Society Meeting, July 14, 2017
1. *Life is hard... to find, but let's look around Saturn for it!* Astronomy on Tap ATX, March 20, 2017.

AWARDED GRANTS

Development of a community data reduction framework to advance EPRV towards the cm/s precision era, NASA ROSES Exoplanet Mass Measurement Program

→ **PI: Daniel Krolkowski**, with 3 Co-Is and 11 Collaborators

→ **Selected**, Funding Amount: \$350,207, Start: March 2026

A community driven, modular data-pipeline architecture to push EPRV into the 1 cm/s era, NASA ROSES Program

→ PI: Jennifer Burt (NASA JPL), 13 Co-Is incl. **Daniel Krolkowski**

→ Total funding: \$479,475, Funding Period: October 2023 – September 2025

TEACHING

Teaching Assistant: Search for Extraterrestrial Life
UT Austin, Astronomy 309L

Spring 2019 & Spring 2020

Undergraduate Lab Instructor: Introductory Astronomy
SUNY Geneseo, Astronomy 101, 106, 111

Fall 2013 - Spring 2016

Physics Learning Center Tutor
SUNY Geneseo

Spring 2014 – Spring 2016

SERVICE, OUTREACH, AND MENTORING

Community Service

- Referee for A&A and AAS Journals
- Panelist for NASA grant reviews (3 times, 2023 – 2025)
- NN-EXPLORE Time Allocation Committee member (4 times, 2024 – 2025)
- Co-founder of Steward Observatory graduate student and postdoc seminar (2023 – present)
- UT Austin graduate student recruitment organizer (2019 & 2020)

Public Outreach

- Co-host of “Astronomy in the News” for Astronomy on Tap Tucson and Austin
- Public lectures at Astronomy on Tap and the Austin Astronomical Society
- Volunteer at outreach event “STEM Girl Day at UT Austin” (2019 & 2020)

Student Mentoring

- Co-research mentor for University of Arizona undergraduate Suhani Surana (Summer 2023)
- Career mentor for UT Austin TAURUS program undergraduates (Summers 2019 & 2021)
- Informal mentor for UT Austin Astronomy program undergraduates (2019 – 2021)

OBSERVING PROPOSALS AND EXPERIENCE

Hobby Eberly Telescope (Habitable-zone Planet Finder)	<i>250+ Hours awarded, 2018 - present</i>
Harlan J. Smith Telescope (Tull Spectrograph)	<i>52 nights (45 PI awarded), 2016 - 2021</i>
WIYN 0.9-m Telescope (Half Degree Imager)	<i>4 nights, 2013-2014</i>
Large Binocular Telescope (PEPSI)	<i>2.5 nights PI awarded, 2026</i>

AWARDS AND DISTINCTIONS

Fred T. Goetting Jr. Memorial Fellowship	<i>September 2021</i>
→ <i>Department of Astronomy, University of Texas at Austin</i>	
Strategic Undergraduate Research Partnership (SURP) Award	<i>September 2020</i>
→ <i>NASA Jet Propulsion Laboratory</i>	
Presidential Scholar	<i>May 2015</i>
→ <i>State University of New York College at Geneseo</i>	
Phi Beta Kappa Honor Society	<i>May 2015</i>
→ <i>State University of New York College at Geneseo</i>	
Sigma Pi Sigma Physics Honor Society	<i>December 2014</i>
→ <i>State University of New York College at Geneseo</i>	

REFERENCES

- Dr. Chad Bender, Astronomer, Steward Observatory**
→ Postdoc advisor and NEID/HPF Instrument Scientist
- Dr. Adam Kraus, Professor of Astronomy, UT Austin**
→ Ph.D. advisor and active collaborator
- Dr. Jennifer Burt, EPRV Investigation Scientist, NASA JPL**
→ PI of EPRV community data standard project and active collaborator